Describing Motion Around Us

Grade 9 Science, Chapter 4: Most Important Questions with Answers

Key Formulas

Average speed	total distance / time interval
Average velocity	displacement / time interval ($v_{av} = s / t$)
Average acceleration	$a = (v - u) / (t_2 - t_1)$
Kinematic equations	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$
Uniform circular motion	$v = 2\pi R / T$ (speed constant, velocity direction changes)
Unit conversion	1 km/h = 5/18 m/s

A. Concept Questions

Q1. Differentiate between distance travelled and displacement.

Ans: Distance is the total path length (scalar, no direction). Displacement is the net change in position between two instants (has magnitude and direction). Magnitude of displacement ≤ distance; they are equal only when the object moves in a straight line without turning back.

Q2. When will the displacement of an athlete running back and forth on a track be zero? What is the distance then?

Ans: Displacement is zero when the athlete returns to the starting point. The distance travelled is then twice the one-way distance (or more), never zero.

Q3. Does fuel used depend on distance travelled or displacement?

Ans: Distance travelled. Fuel is used along the whole path, even if the vehicle ends up back at the start (displacement zero).

Q4. Define average velocity and average acceleration. Can an object move fast and still have zero acceleration?

Ans: Average velocity = displacement / time interval. Average acceleration = change in velocity / time interval. Yes: a bus at constant high velocity has zero acceleration, because acceleration depends on how quickly velocity changes, not on how fast the object moves.

Q5. When is the magnitude of average velocity equal to average speed? When is it zero while speed is not?

Ans: (i) Equal when the object moves in a straight line in one direction without turning back. (ii) Velocity is zero when displacement is zero (object returns to start) while distance is not zero, e.g. one full lap of a track or a swimmer's return trip.

Q6. Girl's scooter has a constant speedometer reading. Can it be accelerating?

Ans: Yes. Constant speed does not mean constant velocity. If the scooter turns (changes direction), its velocity changes, so it is accelerating.

Q7. What does the shape of a position-time graph tell us? What does its slope give?

Ans: A straight sloping line means constant velocity; a curve means changing velocity (accelerated motion); a line parallel to the time axis means the object is at rest. The slope gives velocity.

Q8. What do the slope and the area of a velocity-time graph give?

Ans: Slope = acceleration. Area between the graph and the time axis = displacement. A line parallel to the time axis means constant velocity (zero acceleration); an upward slope means acceleration in the direction of velocity; a downward slope means acceleration opposite to velocity.

Q9. Why is uniform circular motion accelerated even though speed is constant?

Ans: Velocity is along the tangent at each point, so its direction changes continuously. A change in direction means a change in velocity, hence acceleration.

Q10. A student says 'The Earth moves around the Sun'. Can an object kept on Earth be called at rest?

Ans: Rest and motion depend on the reference point. The object is at rest with respect to the Earth (its position does not change), but in motion with respect to the Sun.

Q11. Why is it important to keep a safe distance from the vehicle ahead?

Ans: Stopping distance depends on speed ($s = u^2/2|a|$, so doubling speed makes it four times), road surface, brake capacity and driver reaction time. A larger safe distance is needed at higher speeds.

B. Numerical Problems

Q12. Father goes 250 m to the shop, returns home, goes to the shop again, buys provisions and returns home. Find total distance and displacement.

Ans: Distance = 250 × 4 = 1000 m. Displacement = 0 (he ends at home).

Q13. A student runs from the ground floor to the 4th floor and then comes down to the 2nd floor (each floor 3 m). Find vertical distance and displacement.

Ans: Up: 4 × 3 = 12 m. Down: 2 × 3 = 6 m. Distance = 18 m. Displacement = 12 - 6 = 6 m upward.

Q14. Road trip: 200 km north in 3 h, then 200 km south in 2 h. Find average speed and average velocity.

Ans: Average speed = 400 km / 5 h = 80 km/h. Displacement = 0, so average velocity = 0.

Q15. A car starts from rest and reaches 24 m/s in 6 s. Find acceleration and distance.

Ans: $a = (24 - 0)/6 = 4$ m/s^2. $s = ½ \times 4 \times 6^2 = 72$ m.

Q16. A motorbike with u = 28 m/s stops after 98 m. Find acceleration and time.

Ans: $0 = 28^2 + 2a(98)$ gives $a = -4$ m/s^2. $t = (0 - 28)/(-4) = 7$ s.

Q17. A truck slows from 54 km/h to 36 km/h in 36 s. Find the distance travelled (constant acceleration).

Ans: u = 15 m/s, v = 10 m/s. $s = (u + v)t / 2 = (25 \times 36)/2 = 450$ m.

Q18. A car accelerates from rest to 20 m/s in 5 s, moves at 20 m/s for 10 s, then brakes to rest in 6 s. Find total distance.

Ans: Phase 1: ½ × 20 × 5 = 50 m. Phase 2: 20 × 10 = 200 m. Phase 3: ½ × 20 × 6 = 60 m. Total = **310 m**.

Q19. A bus at 36 km/h sees an obstacle 30 m ahead. Reaction time is 0.5 s; then it brakes with a = 2.5 m/s^2. Will it stop in time?

Ans: u = 10 m/s. Reaction distance = 10 × 0.5 = 5 m. Braking distance = $u^2/2a = 100/5 = 20$ m. Total = 25 m < 30 m, so **yes**, it stops 5 m before the obstacle.

Q20. Brakes give a = -4 m/s^2. Find the stopping distance at (i) 54 km/h and (ii) 108 km/h.

Ans: $s = u^2/8$. (i) u = 15 m/s: s = 28.1 m. (ii) u = 30 m/s: s = 112.5 m. Doubling the speed quadruples the distance.

Q21. Bus: 36 km/h to 54 km/h in 10 s, then braking to rest in 5 s. Find both accelerations.

Ans: (i) a = (15 - 10)/10 = 0.5 m/s^2. (ii) a = (0 - 15)/5 = -3 m/s^2 (opposite to velocity).

Q22. A car moves at 6 m/s for 2 min, then accelerates at 1 m/s^2 for 6 s. Find displacement in 2 min 6 s.

Ans: Phase 1: 6 × 120 = 720 m. Phase 2: 6 × 6 + ½ × 1 × 36 = 36 + 18 = 54 m. Total = **774 m**.

Q23. Car A reaches 5 m/s in 5 s; car B reaches 3 m/s in 10 s (both from rest). Find accelerations and displacements.

Ans: a_A = 1 m/s^2, a_B = 0.3 m/s^2. s_A = ½ × 1 × 25 = 12.5 m. s_B = ½ × 0.3 × 100 = 15 m. (Plot v = at at five instants to draw both lines.)

Q24. Cyclist's v-t graph (Fig. 4.30): 0 to 3 m/s in 20 s, constant 3 m/s until 100 s, then falls to 2 m/s at 120 s. Find displacement and average acceleration for 120 s.

Ans: Area: ½ × 20 × 3 = 30 m; 80 × 3 = 240 m; ½ (3 + 2) × 20 = 50 m. Displacement = **320 m**. Average acceleration = (2 - 0)/120 ≈ 0.017 m/s^2.

Q25. Tip of a 7 cm minute hand from 6:00 PM to 7:30 PM. Find distance, displacement, speed, velocity.

Ans: 1.5 revolutions. Distance = 1.5 × 2π(7) = 21π ≈ 66 cm. The hand ends at the diametrically opposite point, so displacement = 14 cm. Time = 5400 s. Speed ≈ 0.012 cm/s; velocity ≈ 0.0026 cm/s (in the direction of displacement).

C. Graph and Derivation Questions

Q26. Fig. 4.28: A and B start and end at the same positions in 10 s, both moving in one direction. Which statements are correct?

Ans: (i) and (ii) are correct. Same displacement gives equal average velocity; both move without turning back, so distance is also equal, giving equal average speeds.

Q27. Fig. 4.27: Do A and B ever have equal velocity?

Ans: The lines are straight with different slopes (A is steeper), so their velocities are different at all times. They meet only at one point (t = 5 s), which means equal position, not equal velocity.

Q28. Derive $v^2 = u^2 + 2as$.

Ans: From v = u + at, t = (v - u)/a. Substitute in $s = ut + ½at^2$: $s = u(v - u)/a + (v - u)^2/2a = (v^2 - u^2)/2a$, so $v^2 = u^2 + 2as$.

Q29. Using the area of a trapezium on the v-t graph, derive s = ½(u + v)t.

Ans: The area under the line from t = 0 to t is a trapezium with parallel sides u and v and width t. Area = ½ × (sum of parallel sides) × width = ½(u + v)t, which equals displacement s.

Q30. Derive $s = vt - ½at^2$.

Ans: From v = u + at, u = v - at. Put in $s = ut + ½at^2$: $s = (v - at)t + ½at^2 = vt - at^2 + ½at^2 = vt - ½at^2$.